



The problem

Epigenomic data form a three-dimensional tensor across assays, cell types and genomic positions, with tens to hundreds of assays and thousands of cell types. Due to the high cost and complexity of profiling every possible assay in all relevant cell and tissue types, this tensor is sparse: most experiments were never performed. Epigenome imputation methods aim to predict these missing experiments [1].

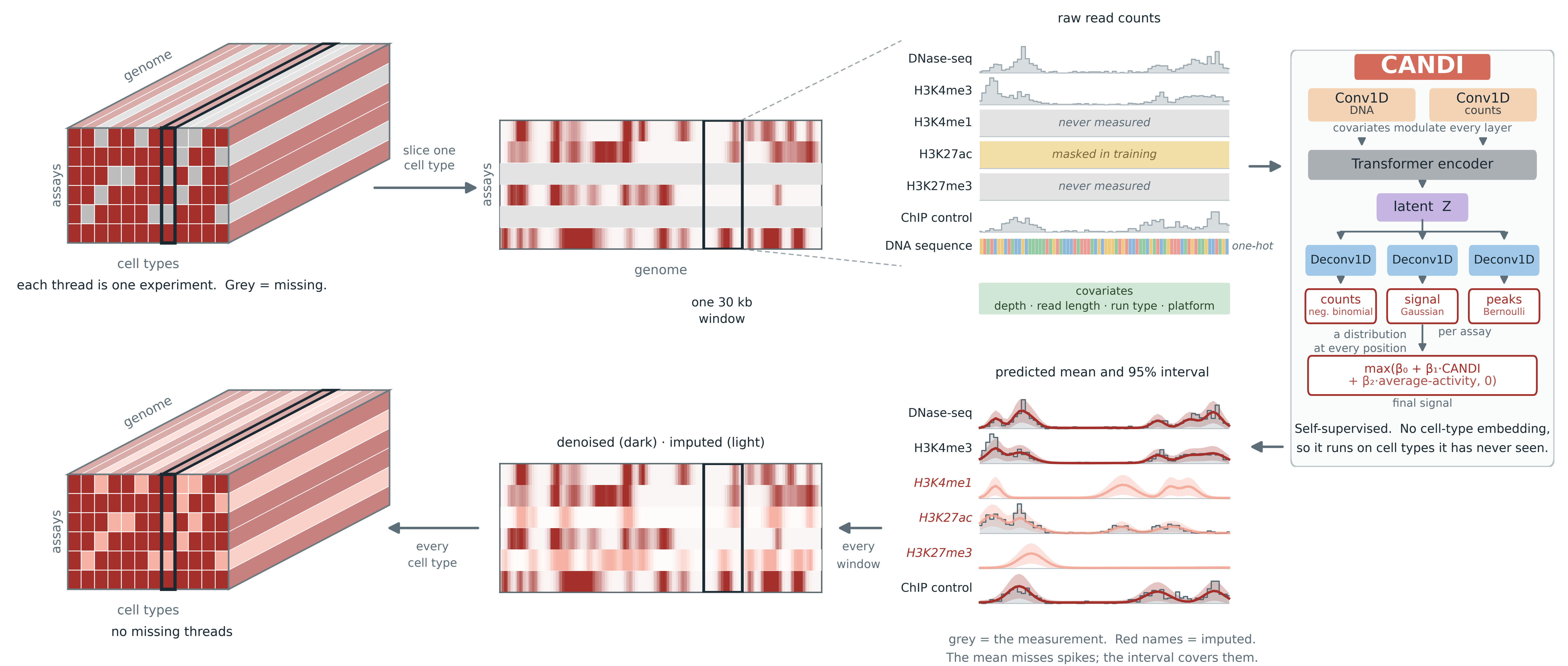
CANDI

Confidence-Aware Neural Denoising Imputer [2] is a self-supervised deep learning method for epigenome imputation and denoising. It takes raw read counts, DNA sequence and experimental covariates, and outputs probability distributions over counts, signal and peaks. Conditioning on metadata rather than cell-type embeddings enables zero-shot generalization to unseen cell types.

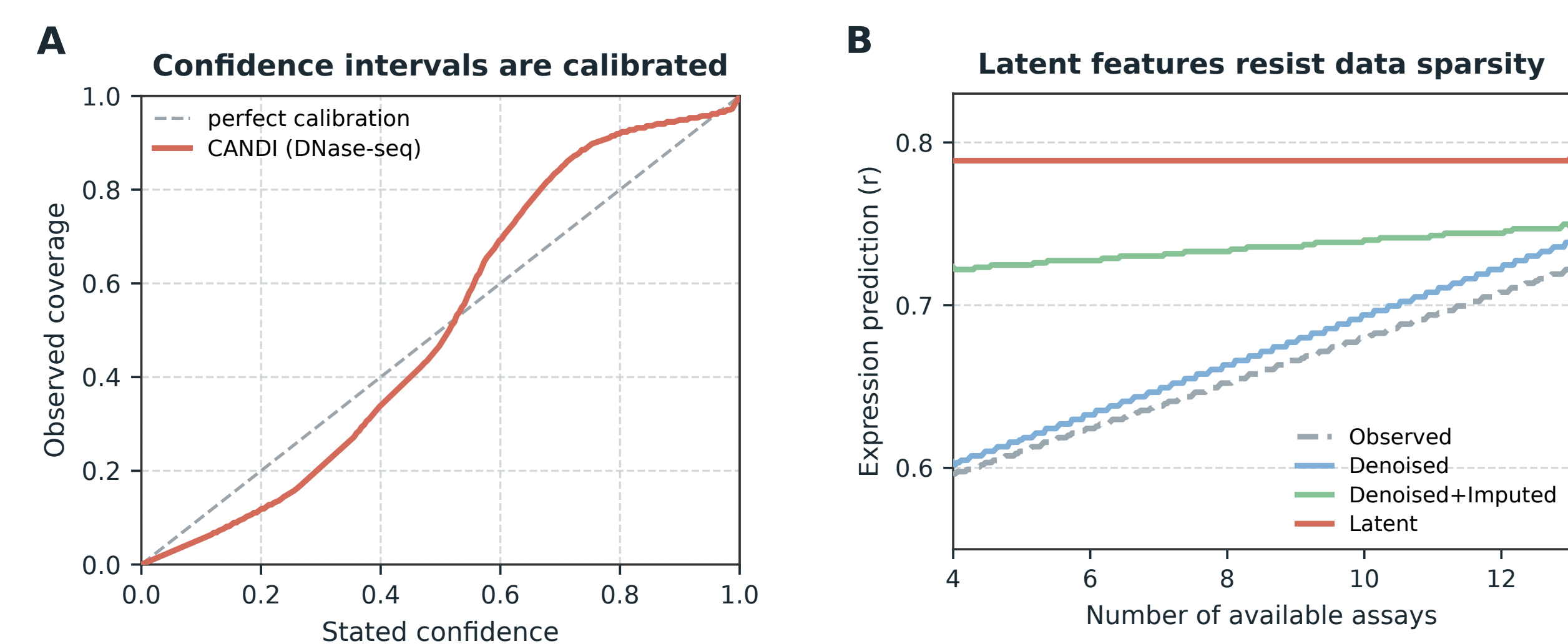
Key findings

- CANDI ranks **1st of 26** on the ENCODE Imputation Challenge.
- CANDI provides calibrated uncertainty estimates.
- CANDI's imputed signals and latent space predict gene expression.

CANDI schematic workflow



Downstream utility of CANDI



A Confidence calibration for DNase-seq (DND-41): predicted confidence level against the fraction of observed data within that interval.

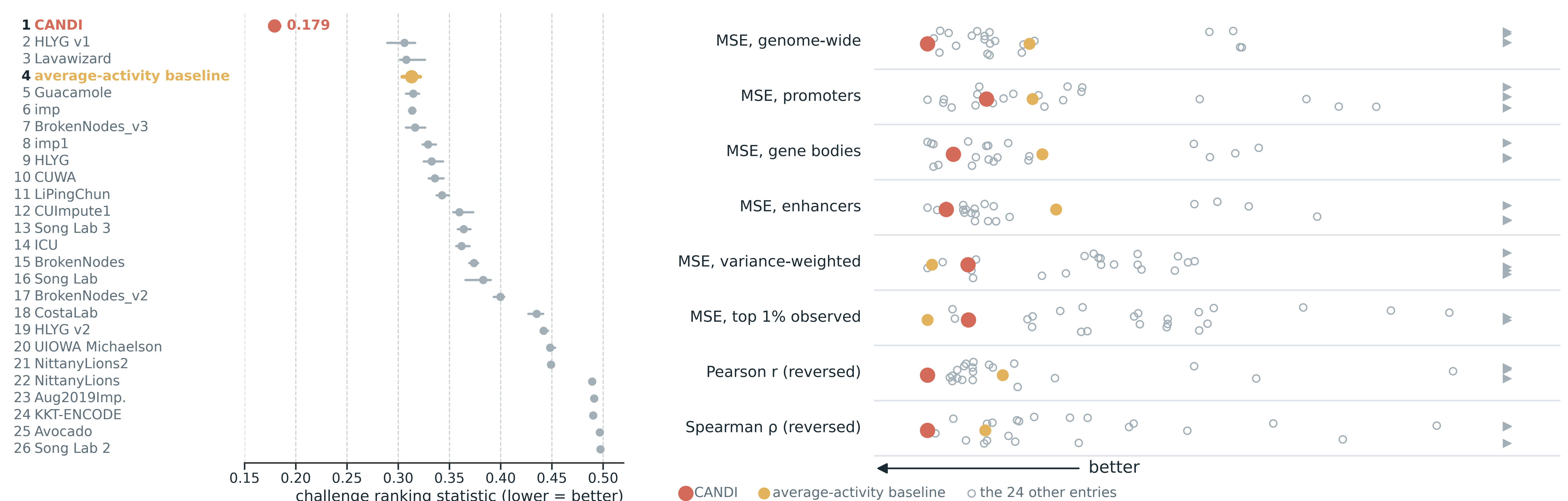
B Gene expression prediction as input assays are removed. Denoised+Imputed and Latent settings are robust to input sparsity.

[1] Schreiber J, Boix C, *et al.* The ENCODE Imputation Challenge: a critical assessment of methods for cross-cell type imputation of epigenomic profiles. *Genome Biol* 24:79 (2023).

[2] Foroozandeh M, Diab AR, Libbrecht M. CANDI: self-supervised, confidence-aware denoising imputation of genomic data. *bioRxiv* 2025.01.23.634626 (2025).

Panels C–D: blind-test scoring from the Libbrecht lab's EIC reproduction; *baseline* is the challenge's average-activity track. The correction was chosen after the challenge results were public; none of it is fitted on blind-test data.

Fixing the train-test experimental covariate shift in the ENCODE Imputation Challenge



C The leaderboard with CANDI added: CANDI is first at 0.179 (bootstrap range 0.177–0.184), the next entry 0.306.

D Eight of the nine measures relative to the baseline, median over 51 experiments. CANDI is first on three and top 8 on the rest.